

Question	Marking Guidance	Mark	Comments
03.1	1. (So the) oxygen is used/absorbed/respired;	1	
03.2	1. <u>Anaerobic</u> respiration produces carbon dioxide; 2. Increase in pressure/volume (of gas);	2	2. Reference to either volume or pressure required for the mark
03.3	1. Correct answer in range of 4.9×10^{-4} to 4.91×10^{-4} = 2 marks ;; 2. Incorrect answer but shows division by 24 = 1 mark OR Incorrect answer but shows a number from 1175 to 1178 (ignore position of decimal point, standard form and any numbers that follow) = 1 mark ; OR Incorrect answer but show the number 49 (ignore position of decimal point, standard form and any numbers after 49) = 1 mark ;	2	1. Accept any equivalent mathematical representation of this answer
03.4	Large range/difference/increase in numbers;	1	1. Accept reference to exponential (increase) 1. Ignore if the answer only refers to numbers being high 1. Ignore to 'fit on the scale'
03.5	1. Decrease/no glucose/substrate OR Increase in ethanol/carbon dioxide/acidity;	1	Accept decrease/no oxygen as Figure 4 is not linked to Figure 3. Accept competition for glucose/oxygen. Accept any named sugar Accept decrease in pH Accept increase in toxins 1. Ignore food/nutrients

03.6	1. Correct answer of 298000 or 297766 or 297765.59 or 296826 = 2 marks;; 2. Incorrect answer but working shows 2000×2.72 = 1 mark; OR Incorrect answer but working shows $2.72^{0.5 \times 10} / 2.72^5 / e^{0.5 \times 10} /$ = 1 mark	2	1. Accept: any equivalent answer with appropriate rounding e.g. 2.98×10^5, 29.78×10^4 etc.
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